

Music and Events Ontology

Group F

Muhammad Umer Raza 12436441

Margarida M. A. Gonçalves 12142151

Ontology & Application Domain

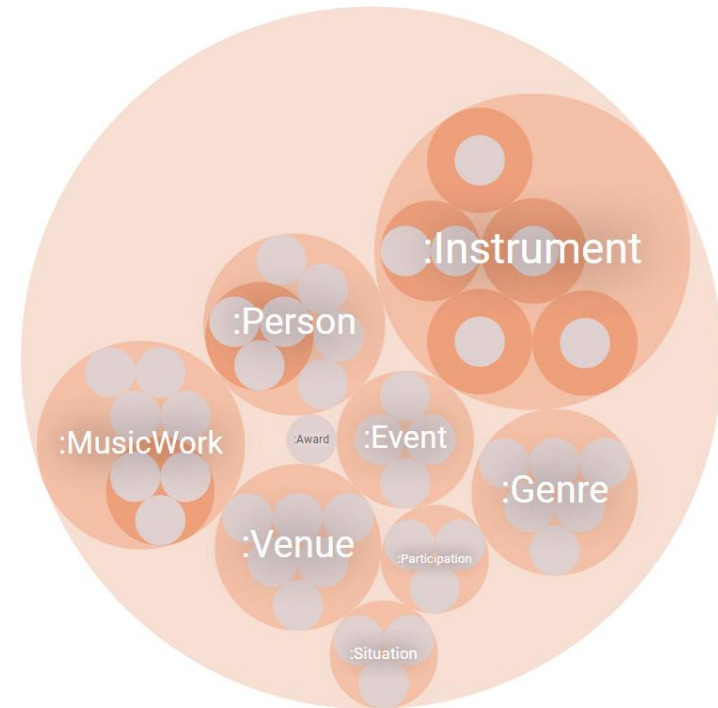
Overview

- **Domain:** Comprehensive ecosystem for music and events such as concerts
- **Goal:** To connect creative intellectual property (albums, tracks, genres) with physical event contexts
- **Competency Question:**
 1. Which artists perform at a given concert?
 2. Which tracks belong to a given album and what genres do they cover?
 3. What is the relationship between a venue's capacity and the actual concert audience?

Ontology & Application Domain

Overview

- **Classes:** 60
- **Properties:** 39 (Object) + 22 (Data)



Step 1 – Advanced Ontology Engineering

Some Advanced OWL 2 Features Used:

- **Disjoint classes:**
 - $\text{Band} \sqcap \text{SoloArtist}$ are disjoint
- **Cardinality restrictions:**
 - Album has at least 1 Track (Existential Restriction)
- **Equivalent classes:**
 - $\text{LiveAlbum} \equiv \text{Album} \sqcap (\text{recordedAt some Concert})$
- **Comments and Meta Data:**
 - `Rdfs:label` & `rdfs:comment`
- **Property chains:**
 - $\text{hasMember} \circ \text{playsInstrument} \rightarrow \text{featuresInstrument}$

Step 1 – Advanced Ontology Engineering

Ontology Design Patterns (ODPs)

- **Participation Pattern:** Participation reifies the artist-concert relationship so multiple artists can hold different roles at the same event (N-ary)

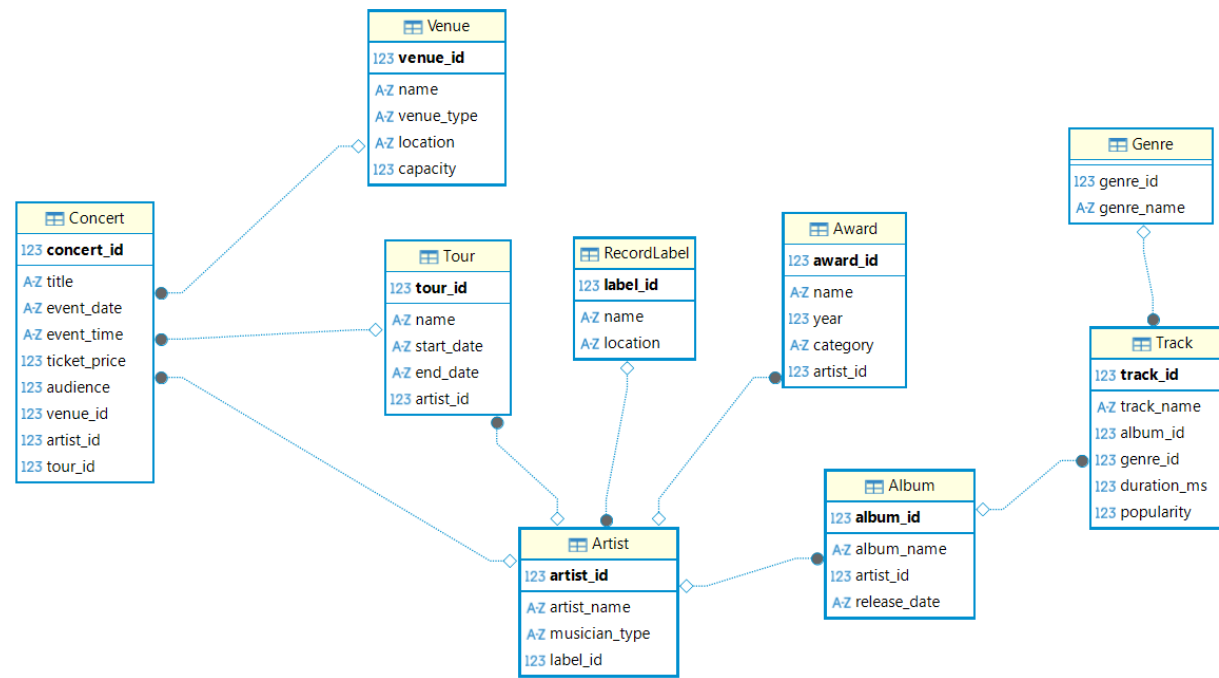


- **Part-Of Pattern:** Reifies a complex part-whole architecture mapping a Concert Program to a structured sequence of Program Items (individual track performances)

Step 2 - KG creation through OBDA

Datasets used

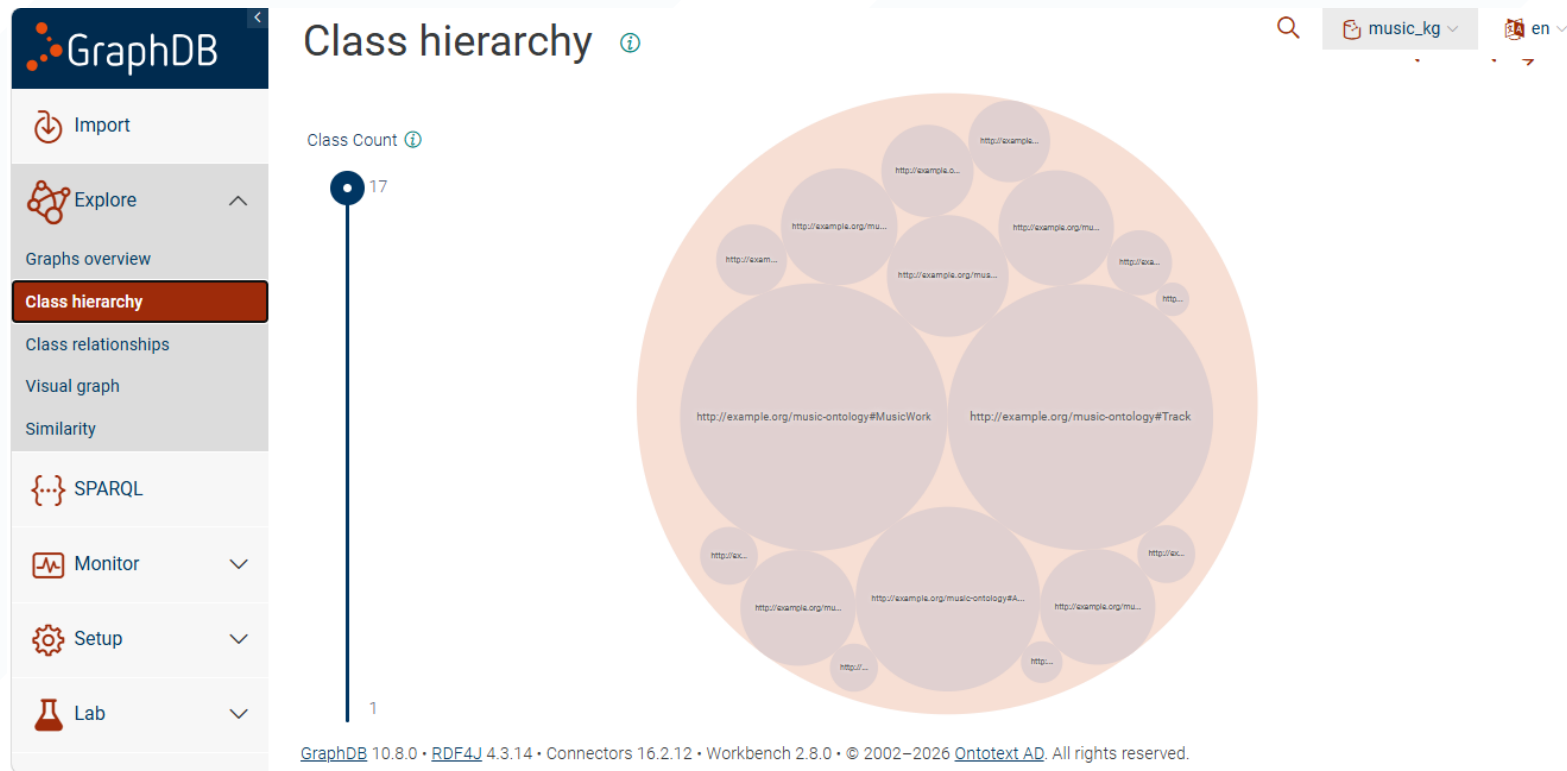
- **Spotify Kaggle Data:** Information about songs, artists and playlists
- **Generated Data:** Venues, concerts and awards



Step 2 - KG creation through OBDA

GraphDB

- R2RML Mapping
- Materialize the KG using Ontop Plugin: Total triples 72,419
- 17 ontology classes were successfully populated with data.



Step 3 – Verification of Semantic Structures

OOPS! Scanner

- Pitfalls were mainly caused by distraction creating the Classes and Properties
- All critical issues were reviewed and corrected where applicable.

Pitfalls detected:

Results for P04: Creating unconnected ontology elements.	2 cases	Minor
Results for P05: Defining wrong inverse relationships.	2 cases	Critical
Results for P08: Missing annotations.	118 cases	Minor
Results for P11: Missing domain or range in properties.	10 cases	Important
Results for P13: Inverse relationships not explicitly declared.	9 cases	Minor
Results for P22: Using different naming conventions in the ontology.	Ontology*	Minor
Results for P29: Defining wrong transitive relationships.	4 cases	Critical
Results for P30: Equivalent classes not explicitly declared. 1 case	1 case	Important
Results for P41: No license declared.	Ontology*	Important

Step 3 – Verification of Semantic Structures

SHACL Rules

- Executed via the TopBraid SHACL engine
- Result: sh:conforms true — 0 violations
- Result: All validation constraints passed successfully.

Shape	Description	Target	Complexity
ArtistShape	Musician must have a name	Musician	Simple
VenueShape	Venue must have a name and integer capacity	Venue	Simple
ConcertShape	Concert must have title, date and price	Concert	Simple
AlbumShape	Album must have a title	Album	Simple
TrackShape	Track must have a title	Track	Simple
GenreShape	Genre must have exactly one name	Genre	Medium
TourShape	Tour must have a name	Tour	Simple
RecordLabelShape	RecordLabel must have a name	RecordLabel	Simple
ConcertAudienceShape	Concert audience count must be positive integer	Concert	Medium
NoSelfGenreShape	Track must not be its own genre	Track	SPARQL

Step 4 - Ontology Alignment

- **Performed Music Ontology (PMO)** using **AgreementMakerLight**
- **3 high-confidence mappings:**
 - RecordingSession 0.99
 - Event 0.98
 - Concert 0.98
- **Strengths:**
 - High confidence on structural matches:
 - Cross-ontology BIBFRAME link enables library catalogue integration
 - Improves interoperability with external ontologies
- **Weaknesses:**
 - Only 3 matches
 - PMO focuses on classical music and ours focuses on contemporary popular music.

Why Only 3 Ontology Matches?

Different ontology objectives

- Our ontology models the **popular music industry** (artists, albums, tours, awards, record labels).
- PMO models **performed music**, recordings, and performance events.

Limited semantic overlap

- Many classes (e.g., **Award**, **Tour**, **RecordLabel**) have no direct equivalent in PMO.

High-confidence matching

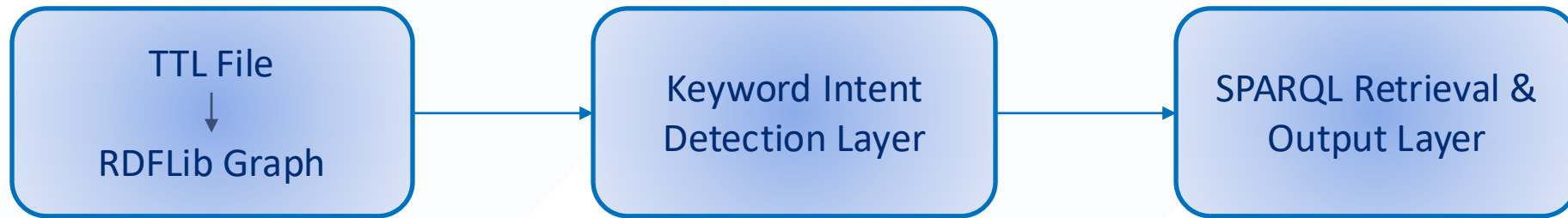
- AgreementMakerLight reports only mappings with strong lexical and structural similarity.
- This avoids incorrect or low-confidence alignments.

Correct matches found

- RecordingSession ↔ PMO:RecordingSession
- Concert ↔ PMO:Concert
- Event ↔ BIBFRAME:Event

Step 5 – KG-based RAG application

Architecture • Implemented using Python, RDFLib and SPARQL queries



- **Sample Q&A:**

"How many artists?" → 50

"List all artists" → The Chainsmokers, Martin Garrix, Queen...

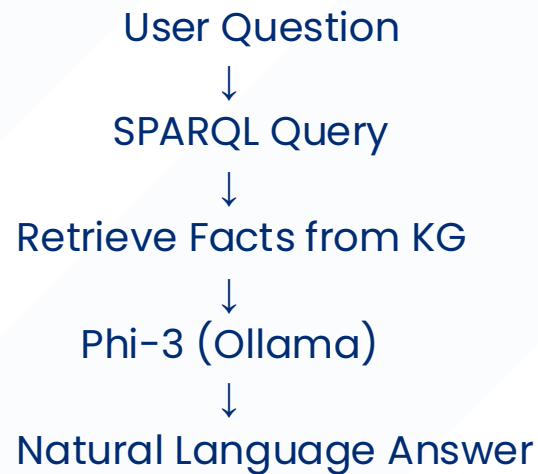
"What classes exist?" → Musician, Album, Track...

Step 5.1 – LLM-based Answer Generation

Local LLM Integration

- **LLM Used:** Microsoft **Phi-3**
- **Inference Engine:** **Ollama**
- **Implementation:** `rag_llm.py`
- **Knowledge Source:** Materialized Music Knowledge Graph (RDFLib + SPARQL)

Question Answering Workflow



Challenges & Limitations

Challenges

- One group member dropped out during the project.
- Only **3 high-confidence ontology mappings** were found because PMO and our ontology model different aspects of the music domain.
- Question understanding is based on **keyword matching**, limiting flexibility for complex queries.

Current Limitations

- Supports mainly **single-hop SPARQL** queries.
- Answer quality depends on the completeness of the Knowledge Graph.
- Limited support for complex reasoning across multiple entities.

Future Work

- Support **multi-hop semantic reasoning**.
- Improve intent detection using semantic retrieval.
- Extend ontology alignment with additional music ontologies.
- Develop a web-based interface for easier interaction.

Thank You